

Related work: digital mice in context

Executive summary

The closest antecedents to “digital mice” fall into six threads. First, recent work explicitly reframes cognitive bias and bounded rationality as *instruments* for computational social science rather than defects to mitigate (CoBRA; CoBRA, 2025), and a parallel line uses LLMs as “Homo silicus” — prompt-conditioned implicit models of humans to be experimented on (Horton, 2023; Manning et al., 2024). Second, the silicon-sampling tradition (Argyle et al., 2023) and its digital-twin descendants (Peng et al., 2025) build synthetic populations through demographic persona conditioning, but typically without manipulating reasoning capacity. Third, an emerging auctions-and-games literature deploys LLMs as economic agents and benchmarks their behavior against the canonical human experimental record (Shah et al., 2025; Zheng et al., 2025; Henning et al., 2025), in some cases formalizing bounded rationality through QRE (Pechon-Elkins and Chun, 2026) or prompt-level equilibrium concepts (Zhu, 2025). Fourth, Li (2017)-style OSP theory (Li, 2017; Breitmoser and Schweighofer-Kodritsch, 2022) provides the diagnostic vocabulary — obvious dominance, contingent reasoning failures — that the digital-mice fitness function reuses. Fifth, multiple empirical studies document that off-the-shelf LLMs are *too* rational, too homogeneous, and too fair relative to humans (Henning et al., 2025; Peng et al., 2025), which is precisely the gap a controlled-deficit pipeline is built to close. Sixth, and methodologically closest, an AI-safety literature already studies how to make a strong model *play dumb*: sandbagging and password-locking deliberately suppress a capable model’s competence (van der Weij et al., 2024; Greenblatt et al., 2024), and “model organisms” induce minimal controlled failure modes as experimental systems (Turner et al., 2025). Digital mice borrow this play-dumb move but repurpose it from a safety threat or interpretability probe into a social-science instrument.

Digital mice are novel along two axes that no prior line covers jointly: (i) systematic, prompt-induced reasoning-capacity knockouts treated as an experimental design variable (a “knockout mouse” for LLM agents), and (ii) quantitative calibration of those knockouts against Li (2017) auction moments, then used as the fitness signal in an evolutionary search over auction genotypes. Prior work has the ingredients — persona conditioning, bias toolkits, auction benchmarks, QRE calibration — but none combines instructed cognitive deficits with moment-matched human auction data inside a mechanism-design search loop.

1. Bias and bounded rationality as research instruments

A small but growing literature reframes LLM cognitive limitations as something to *program in*, not engineer away. CoBRA (CoBRA, 2025) provides a toolkit with a Cognitive Bias Index and a Behavioral Regulation Engine that explicitly controls biases in LLM-based social agents, using validated social-science experiments as the scaffold, and positions “bias not as a flaw but as an instrument for scientific reproducibility and behavioral modeling.” This is the same instrumental move the digital-mice analogy makes: treat the deficit as the apparatus, not the noise. The contrast is along the deficit axis — CoBRA targets classical cognitive biases (anchoring, framing, etc.), whereas digital mice target *reasoning depth and contingent reasoning capacity* relevant to auction strategy. Neither library calibrates against quantitative auction moments.

Zheng et al. (2025) push in the same direction empirically: across repeated games (RPS, Prisoner’s Dilemma) LLMs reproduce familiar human heuristics such as outcome-based strategy switching and increased cooperation under repeated interaction, but “apply these rules more

rigidly and demonstrate weaker sensitivity to the dynamic changes in the game environment.” They conclude that current LLMs “capture only a partial form of human-like bounded rationality,” motivating training methods for flexible opponent modeling. Digital mice take the opposite engineering route: rather than retraining for more flexible bounded rationality, we *prompt-instrument* specific deficits and treat the resulting behavior as the unit of analysis.

2. Homo silicus and automated social science

Horton (2023) introduces the “Homo silicus” framing: LLMs as “implicit computational models of humans . . . they can be given endowments, information, preferences, and so on, and then their behavior can be explored in scenarios via simulation.” Methodologically, the paper replicates Charness–Rabin, Kahneman et al., Samuelson–Zeckhauser, Oprea, and Horton himself, and treats *qualitative* agreement (or generative divergence) as the validation criterion. Digital mice keep the prompt-as-endowment idea but replace qualitative replication with quantitative moment matching against Li (2017).

Manning et al. (2024) push this to the full research loop: LLMs as both *scientists* (hypothesis generators) and *subjects* (experimental agents), with structural causal models serving as the language for stating hypotheses, building agents, designing experiments, and analyzing data, demonstrated across negotiations, bail hearings, job interviews, and auctions. Crucially, the paper does not introduce controlled cognitive deficits or bounded-rationality constraints — agents are treated as default reasoners. Digital mice can be read as a specialization of this framework where the SCM “treatment” variable is reasoning capacity itself, instrumented via prompted CoT suppression.

3. Synthetic populations and digital twins

Argyle et al. (2023) coin “silicon sampling” and the criterion of “algorithmic fidelity”: conditioning GPT-3 on thousands of sociodemographic backstories from real survey respondents, with the goal that “proper conditioning will cause it to accurately emulate response distributions from a wide variety of human subgroups.” The construct propagates downstream into the digital-twin literature. Methodologically, this is persona/demographic conditioning, with no explicit treatment of bounded rationality or reasoning-capacity ablation — a clear contrast with the digital-mice approach, which holds persona fixed and varies cognitive capacity.

Peng et al. (2025) stress-test the digital-twin program across nineteen pre-registered replications (164 outcomes) and report an average correlation of only $r \approx 0.20$ with human responses, with documented distortions including a “Targeting Fairness” regime in which LLMs produce systematically skewed fairness judgments in economic and social scenarios. The finding is directly relevant to LLMs-as-bidders or LLMs-as-matching-market participants, where fairness and envy considerations shape bid shading and preference reports. Digital mice respond to this critique by refusing the naive “proxy” framing: rather than asking whether LLMs *are* humans, we ask whether instructed-deficit LLMs can reproduce the *moments* of human play under specific cognitive constraints.

4. LLMs as auction and game-theoretic agents

Shah et al. (2025) deploys LLMs as bidders in canonical experimental auction frameworks (first-price, second-price, and OSP variants), running over a thousand auctions and reporting that LLM bidders “produce results consistent with risk-averse human bidders” and “succumb to the winner’s curse in common value settings.” The paper explicitly engages Li (2017) on obviously strategy-proof mechanisms and Breitmoser and Schweighofer-Kodritsch (2022) on “obviousness” as a theoretical lens for LLM cognitive limitations, and quantitatively benchmarks against Coppinger et al. (1980), Cox et al. (1988), and Kagel (1993) — e.g., matching “the 27% reported by Kagel (1993)” for value-bidding frequency in SPSB. This is the most direct precedent for the auction-side empirical loop in digital mice, sharing both the OSP diagnostic language and the human-baseline calibration target. The key difference: Shah et al. (2025) does *not* systematically manipulate reasoning capacity to recover specific moments, and does not run an evolutionary search over auction formats conditional on agent cognition.

Pechon-Elkins and Chun (2026) provide a complementary calibration tool by estimating Quantal Response Equilibrium λ parameters for LLM agents on a continuous scale against human data (with human $\lambda \in [1.0, 2.5]$ and frontier-model estimates in $[0.05, 1.10]$ across 1,855 games and seven models). QRE gives a principled one-dimensional handle on strategic sophistication, and is in principle complementary to digital mice: QRE measures *how* rational an LLM is, while digital mice manipulate that quantity through instructed deficits.

Zhu (2025) formalizes an “LLM-Nash reasoning equilibrium” where agents choose *reasoning prompts* rather than actions, and defines mindsets as triples (information intake, prompt space, worldview parameters), with equilibrium defined over the prompt space and actions emerging as LLM inference output. This provides a clean theoretical vocabulary for what digital mice do mechanically: deficit-injecting prompts are precisely the strategic objects in $\mathcal{X}, \mathcal{Y}$ above, and the equilibrium concept formalizes why prompt-level manipulations produce behavior distinct from classical Nash play.

5. Out-of-the-box LLMs are too rational and too homogeneous

A robust empirical finding across studies is that frontier LLMs, absent intervention, are systematically *more* rational and *more* homogeneous than human subjects. Henning et al. (2025) run the Smith–Suchanek–Williams bubble paradigm with LLM populations and find that LLMs “generally exhibit a *textbook-rational* approach, pricing the asset near its fundamental value, and show only a muted tendency toward bubble formation,” and that LLM-based agents “display less trading strategy variance in contrast to humans.” They warn that “LLM-only data . . . to replicate human-driven market phenomena” is unreliable because “key behavioral features, such as large emergent bubbles, were not robustly reproduced.” Together with the fairness distortions documented by Peng et al. (2025) and the rigidity findings of Zheng et al. (2025), this body of evidence motivates the central digital-mice design choice: do not assume a baseline LLM matches a human bidder; instead, calibrate a *cognitive treatment* that reproduces specified moments of human play, and use that treated population as the unit of analysis.

6. Researcher-controlled deficit induction: making a strong model “play dumb” on purpose

The thread closest to the digital-mice *mechanism*—taking a capable reasoning model and deliberately degrading its competence in a controlled way—divides cleanly into two classes that should not be conflated. (A) *Model-driven evasion*: the model itself, or a developer with an incentive to pass an evaluation, hides capability. (B) *Researcher-driven controlled induction*: an experimenter systematically dials a model’s competence down as a designed treatment, to study or use the resulting deficient agent. Digital mice are squarely in class (B). We review (A) only to delimit it, then survey the (B) methods that are the true methodological neighbors.

Delimiting the evasion case (A). Sandbagging is the canonical (A) example: van der Weij et al. (2024) define it as “strategic underperformance on an evaluation” and show frontier models can be prompted to “selectively underperform on dangerous capability evaluations” while passing general ones, and that “capable password-locked models can reasonably emulate less capable models.” Greenblatt et al. (2024) train password-locked models that “imitate a much weaker LLM” unless a password is present. These papers share the digital-mice *technique*—a strong model made to act weak—but their *object of study* is the threat that a model or developer evades oversight; the suppression is in service of deception, not measurement. We borrow the one transferable empirical lesson—suppressed capability is fragile and can leak back “from a few high-quality demonstrations”—because it is exactly what the digital-mice reasoning-chain audit quantifies (the 0.001 vs. 0.41 dominance-invocation rate under CoT vs. instruction-only suppression).

Controlled depth limitation by prompt (B). The closest (B) method to the digital-mice contingent/belief axes is level- k and cognitive-hierarchy prompting. Zhang et al. (2024) build a “K-Level Reasoning” framework in which the experimenter sets the strategic depth—reason as a level-0, level-1, . . . , level- k player forming “higher order beliefs, beliefs about others’ beliefs”—and the cognitive-hierarchy benchmark (CHBench, 2025) evaluates LLMs across that explicit depth ladder. This is researcher-controlled bounded rationality of exactly the kind digital mice impose, drawn from the same level- k / cognitive-hierarchy tradition. The difference is direction of use: K-level reasoning raises depth to *improve* strategic play, whereas digital mice hold depth low to *reproduce* a human deficit and then calibrate it to auction moments.

Controlled compute/reasoning-budget limitation (B). A second (B) method limits how much a model is allowed to reason. The test-time-compute literature (Reasoning-on-a-Budget survey, 2025) controls a model’s reasoning under a fixed budget—e.g. BudgetThinker (BudgetThinker, 2025) inserts control tokens that tell the model its remaining token budget, and production systems expose a “thinking budget” knob. Cutting the budget is a continuous, researcher-set deficit dial. Digital mice instead constrain *which reasoning operations* are permitted (contingent, forward, belief) rather than *how many tokens* are spent, but both treat reasoning capacity as a controllable experimental quantity.

Controlled removal at the weight/representation level (B). Where digital mice ablate by prompt, machine unlearning ablates by weights. RMU (Li et al., 2024) “reduces model performance” on a target capability “while maintaining general capabilities,” by redirecting the internal representation of the forgetting set—a surgical, researcher-controlled capability knockout.

Activation steering and persona vectors (Chen et al., 2025) achieve a similar controlled effect at inference time: behaviors and traits correspond to “approximately linear directions in activation space,” and adding or *subtracting* along a direction reliably steers (or suppresses) behavior “without changing the surface prompt.” These are the weight- and activation-level analogues of the digital-mice prompt-level deficit; they show the controlled-deficit philosophy is not specific to prompting.

Model organisms: the exact framing twin (B). Finally, the alignment literature’s *model organisms* are the precise methodological twin of the knockout-mouse framing: deliberately induced, minimal, reproducible failure modes used as controlled experimental systems. Turner et al. (2025) create model organisms of emergent misalignment using “a single rank-1 LoRA adapter,” isolating “a single linear direction that induces” the behavior so defensive techniques can be benchmarked against a known ground truth. The shared logic is exact: induce a controlled deficit, verify it is the intended one, use the resulting strain to stress-test something downstream. Digital mice differ only in the downstream target (a mechanism’s SMAD against human auction data, not an interpretability method) and the induction channel (prompt-level CoT constraints, not weight-level edits).

What is distinctive about the digital-mice variant of (B). Across the (B) methods above, the deficit is induced for a *model-facing* purpose: better strategic play (level- k), cheaper inference (budget control), safety (unlearning), or interpretability (model organisms). None calibrates the induced deficit to a *human behavioral target*. Digital mice are the only (B) instance we found that (i) induces the deficit specifically to match human economic-experiment moments and (ii) then uses the calibrated deficient population as the evaluator in a mechanism search.

7. Where digital mice are novel vs. reuse

Reused ingredients. The instrumental framing of bias/bounded rationality (CoBRA, 2025); the Homo-silicus prompt-as-endowment paradigm (Horton, 2023; Manning et al., 2024); the LLM-as-auction-bidder setup and OSP diagnostic vocabulary (Shah et al., 2025; Li, 2017; Breitmoser and Schweighofer-Kodritsch, 2022); the human-benchmark targets from classical experimental auctions (Coppinger et al., 1980; Cox et al., 1988; Kagel, 1993); the quantitative measurement of LLM bounded rationality via QRE (Pechon-Elkins and Chun, 2026) or prompt-space equilibrium (Zhu, 2025); and—most directly—the practice of making a capable model emulate a weaker one, established as sandbagging / password-locking (van der Weij et al., 2024; Greenblatt et al., 2024) and as controlled deficit induction in model organisms (Turner et al., 2025).

Novel combination. Digital mice (i) treat reasoning-capacity manipulation (CoT suppression, instructed deficits) as a designed experimental variable in the spirit of a knockout mouse—the same “play-dumb” move as sandbagging (van der Weij et al., 2024) and model organisms (Turner et al., 2025), but turned from a safety *threat* or interpretability *probe* into a social-science *instrument*; (ii) calibrate the induced-deficit population against Li (2017) auction moments rather than against qualitative replication (Horton, 2023), single-dimensional QRE (Pechon-Elkins and Chun, 2026), or a generic weaker-model target (Greenblatt et al., 2024); and (iii) plug the calibrated population into an evolutionary search over auction genotypes with an SMAD-only fitness function — a mechanism-design loop with no antecedent in any of the threads above. No prior line combines deliberate

capability suppression, calibration to human economic-experiment moments, and mechanism search; that triple is what is new.

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